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RESEARCH ARTICLE

Addressing Adversarial Attacks in IoT Using Deep Learning AI Models

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ABSTRACT Adversarial attacks, specialized attacks, pose a severe threat to AI model performance in various applications, including the Internet of Things (IoT). Various defense mechanisms have been proposed to counter these attacks. However, their primary limitation lies in their inability to effectively handle broader datasets derived from diverse applications. In this study, we utilize multiple AI models with adaptive weights applied at different neural network layers to achieve enhanced performance and more robust results. This study introduces a novel AI-based deep learning model to detect adversarial threats within IoT systems, optimizing data preprocessing, feature extraction, and classification through a holistic approach. A three-stage filtering technique featuring Adaptive Weights was applied to enhance the data preprocessing efficiency. A two-level adaptive feature extraction strategy was utilized to maximize feature learning performance. This is refined using adaptive dilated enriched convolution operations, whereas statistical attributes are optimized through a Quantum-inspired Coati Optimization Algorithm (Q-COA). A dual system based on self-attention combines a Restricted Boltzmann Machine (RBM) with a Recurrent Convolutional Neural Network (RCNN). This configuration effectively identifies adversarial attacks by linking classifiers via a self-attention-driven weight-sharing mechanism. The proposed two-level weight-sharing approach surpasses conventional classifiers and achieves superior classification accuracy. This comprehensive Artificial Intelligence (AI) model significantly improves the preprocessing efficiency, feature learning performance, and classification accuracy, offering an innovative and robust solution for adversarial attack detection in IoT systems. The performance metric, Area Under the Curve (AUC), achieves values of 0.95 and 0.97 for two datasets using the proposed model, highlighting its effectiveness compared to the models in the comparison.

INDEX TERMS Adversarial attacks, IoT, CNN.

I. INTRODUCTION

The Internet of Things (IoT) has become vital for intelligent applications such as smart homes [1]. IoT consists of many physical devices or sensors that gather and exchange data from their surroundings while connected to the Internet. Edge AI in IoT devices refers to processing capabilities directly on local devices at the “edge” of a network rather than relying on centralized cloud-based servers. This enables data processing

and decision-making to occur closer to the data source. IoT gateways receive data from small, low-power devices with limited computing resources, aggregate them, and transmit them to Internet-based networks. Security has become an increasingly critical concern in the IoT industry [1]. The main challenge stems from the diversity of devices, communication methods, architectures, and enormous data transmission across networks [2], [3].

As shown in Fig. 1, a typical IoT environment consists of numerous sensors connected to edge devices, which are further linked to the cloud via the Internet across various

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geographical locations. These sensors generate large volumes of data, which are transmitted to edge devices and subsequently to the cloud through unsecured channels, such as the Internet. This data is highly vulnerable to manipulation by intruders at any stage of the IoT ecosystem. Such manipulation can lead to device malfunctions resulting from incorrect decisions made by the processing modules. This form of data manipulation is known as an adversarial attack.

An adversarial attack is a security breach that can lead to misclassification or system malfunction. These attacks involve manipulating an IoT system's input data in ways that cause the system to make inaccurate decisions [4], [5]. This can be achieved by introducing subtle perturbations to the input data, which are invisible to humans but can cause the system to classify the input incorrectly [6].

IoT systems are particularly vulnerable to adversarial attacks because they can cause physical damage or disrupt assets. One of the most critical challenges in IoT security is the increasing threat of these attacks, where malicious actors exploit weaknesses to compromise the confidentiality, availability, and integrity of IoT devices and systems [7], [8].

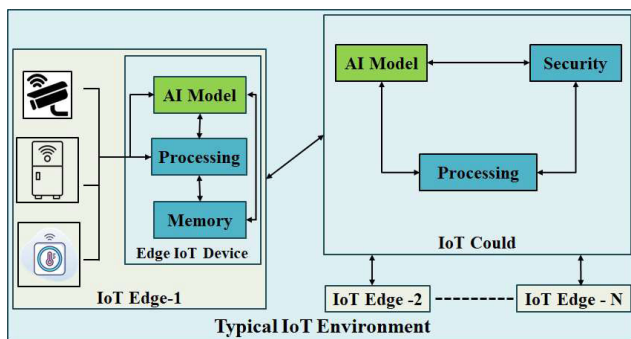


FIGURE 1. An AI model usage in IoT edge and cloud to address adversarial attacks.

IoT security research focuses on defending against malicious attacks, as traditional security measures often struggle to keep pace with the evolving threat landscape. Reference [9] highlighted the vulnerabilities of IoT systems to adversarial inputs, emphasizing the need for security measures that can operate within the constraints of IoT hardware. Current research often relies on a single AI model [9], which is problematic because no single approach can effectively address all attack scenarios.

This highlights the need for innovative solutions to bolster the security of IoT systems [7]. Adversarial training and statistical methods would help reduce the vulnerabilities. As illustrated in Fig. 1, the AI model is pre-trained on extensive datasets and operates in real time to detect malicious attacks while filtering out noise from the data. The model can be deployed at both the edge and cloud levels, enabling noise filtration at the device level and further data refinement in the cloud. This dual-layer approach ensures that accurate processing decisions are made in real time.

To address the limitations of current research [9], this paper proposes a heterogeneous methodology that creates a multi-layered [10], resilient system that leverages the strengths of different approaches, mitigates individual weaknesses, and increases the overall detection effectiveness against adversarial attacks. This paper proposes a two-tier feature learning approach with a dual classification framework for detecting and defending against adversarial attacks in IoT environments. Effective data filtration is essential because adversarial attacks manipulate the data used by AI models. An optimal classifier is also required for accurate classification. Given the importance of filtering noise from input data and achieving optimal classification, the primary contributions of this study are as follows:

- 1) A three-stage filtering technique was applied for highly efficient data preprocessing, incorporating Adaptive Weighted Gaussian Filtering, Adaptive Weighted Mean Filtering, and Adaptive Wavelet Filtering to clean and transform data before being input into the AI algorithm.
- 2) A two-level feature extraction strategy optimizes the feature learning performance. This involves a deep learning approach with an adaptive feature extraction strategy that integrates Weighted Adaptive Batch Normalization, achieved through a self-adaptive Quantum-inspired Coati Optimization Algorithm (Q-COA).
- 3) For enhanced classification, a robust dual classification system is implemented, combining self-attention with a two-tier classification layer structure that leverages a Recurrent Convolutional Neural Network (RCNN) alongside a Restricted Boltzmann Machine (RBM).

The proposed methodology combines various filtering techniques and integrates classifiers with an adaptive weight-sharing approach to filter out noise, extract features, and enhance classification performance. As Section II outlines, previous studies have proposed various solutions to address adversarial attacks, often relying on a single model or dataset with limited parameters. In contrast, this methodology leverages the strengths of multiple models and a comprehensive dataset encompassing both IoT and IIoT environments, thereby increasing robustness. This approach offers several advantages over single-model defenses and existing methods. By incorporating diversity, redundancy, and complementary defense mechanisms, it strengthens resilience against adversarial attacks. Although not completely invulnerable, it outperforms many single-model defenses, making it significantly more challenging for attackers to succeed.

The remainder of this paper is structured as follows: Section II reviews relevant prior studies, and Section III presents the proposed technique. Section IV details the findings, and Section V concludes the study with a summary and recommendations for future research.

II. LITERATURE REVIEW

This study explores the most advanced methods currently employed to protect IoT devices, aiming to identify gaps,

challenges, and potential research directions for building secure and robust IoT ecosystems. Table 1 summarizes critical studies on adversarial attacks and defense mechanisms in AI-based IoT security.

Ibitoye et al. [11] discuss the vulnerability of IoT systems to adversarial attacks and the challenges they pose to intrusion detection systems (IDS). It highlights the increasing reliance on machine learning in IoT security and how adversarial attacks exploit model weaknesses. The abstract underscores the need for effective defense mechanisms and analyzes adversarial threats in IoT environments. The study emphasizes the importance of enhancing IDS to withstand these threats, proposing techniques to improve their robustness and reliability. Self-normalizing Neural Network (SNN) was used in this proposal. The paper does not explore how well the proposed solutions generalize across this diversity.

Cakir et al. [12] proposed a Gated Recurrent Unit (GRU) network model based on deep learning to predict and mitigate high-frequency (HF) attacks on the (RPL) protocol in IoT networks. The model's performance was compared with that of a Support Vector Machine (SVM) and logistic regression techniques after analyzing and experimenting with various power states and the overall energy consumption of the nodes. Once the characteristic threshold value is confirmed, the attack prevention mechanism removes any packet from a node that falls outside the specified range. However, this approach faces scalability challenges.

Nayak et al. [13] introduced a Generative Adversarial network classifier (GAN-C), which is a hybrid two-stage model that combines GAN and a Support Vector Machine (SVM) for detecting attack events. The study found that GAN-C outperformed the standalone SVM classifiers. Using a parallel learning and detection paradigm, the proposed method enabled the deep learning of IoT devices with limited processing power. The evaluation results compared the effectiveness of distributed and centralized attack detection in a routing protocol for low-power and lossy network (RPL) IoT networks. However, the model was not updated by using real-time data.

Fu et al. [14] explored the resilience of three intrusion detection models using the Fast Gradient Sign Method (FGSM) to create adversarial scenarios. Models based on Convolutional Neural Networks (CNN), long short-term memory (LSTM), and Gated Recurrent Units (GRU) were evaluated for their ability to detect intrusions in IoT environments. This research demonstrated the potential of advanced deep learning algorithms to offer practical, cost-effective solutions for identifying and mitigating risks in IoT systems. However, this method weakens the decision boundaries of typical instances, slightly compromising the accuracy of routine case detection.

Sahu et al. [15] proposed a novel security architecture and attack detection method that effectively identifies malicious devices using deep-learning models. This method extracts

TABLE 1. An overview of the relevant research on DL-based IoT security.

Author & Year	Methodology	Features	Challenges
[11] Ibitoye et al. 2019	SNN	SNN and compare its performance with the FNN for classifying intrusion attacks in an IoT network	The paper does not explore how well the proposed solutions generalize across this diversity
[12] Cakir et al. 2020	DL based GRU	The high accuracy rate of HF identification of assaults may impact attack detection and prevention strategies in other IoT ecosystems.	This paper did not explore the system's resilience to adaptive attacks where adversaries evolve their strategies
[13] Nayak, et al. 2021	GAN-C	Delivered a distributed IIoT platform that uses a parallel GAN-C based attack detection model to monitor attack events in real-time.	It will not enable the model to be updated with diverse and real-time data. Relies on one model and one data set.
[14] Fu, et al. 2021	FGSM	Three different adversarial training techniques were used on three models to increase their resilience.	It somewhat impairs the judgment of regular instances by causing the decision border of such examples to become brittle. Relies on one model and one data set.
[15] Sahu et al. 2021	Hybrid DL using CNN and LSTM	IoT assaults from over 30 million traffic flows were analyzed dynamically using a dataset.	Single dataset.
[16] Javeed et al. 2021	Cu-DNNGRU + Cu-BLSTM	A highly affordable and scalable DL with SDN support has been brought forward to detect risks in IoT.	The reliance on SDN controllers introduces a potential single point of failure, which attackers could exploit.
[17] H. Benaddi et al. 2022	GAN	GAN to enhance the training process by handling the unbalanced data	The absence of real-world datasets and benchmarks limits the generalizability and practicality of the findings
[18] Rihan et al. 2023	CNN, RNN, GRU, and LSTM	The suggested ensemble feature selection approach is used to select features for DL models that are developed, trained, and intended to identify IoT dangers.	The study does not address the potential challenges of handling imbalanced datasets, which could affect detection accuracy for rare attack types.
Proposed Work	Adaptive noise filtering, Q-COA, and Self-Attention with two-tier classification in RCNN and RBM.	A heterogeneous approach that employs multiple noise filters, Q-COA feature extraction, and a self-attention-based two-tier classification mechanism to detect and mitigate adversarial attacks. This method is adaptable to a wide range of applications beyond IoT environments. Furthermore, it can be implemented at both edge and cloud layers within the IoT ecosystem, leveraging recent advancements in computational capabilities of edge IoT devices alongside improved power management	Increased computing resources are necessary due to multiple filters and compute-intensive algorithms; however, recent edge IoT device advancements offer more processing power. This challenge can also be addressed by training the AI model in the cloud and performing inference directly on the IoT devices at the edge.

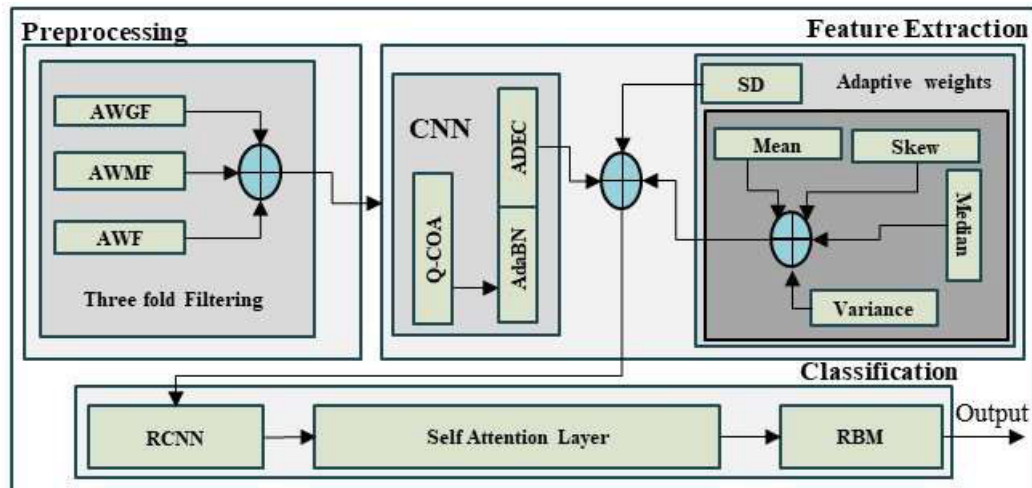


FIGURE 2. Architecture of proposed model connectivity between the phases of the model.

precise feature representations with a CNN and classifies them using an LSTM model. However, the dataset used in this study was not comprehensive.

Javeed et al. [16] introduced a deep learning-driven system with Software-Defined Networking (SDN) support to detect risks in IoT environments. This system uses advanced classifiers for efficient threat detection. The proposed architecture is designed to be highly scalable and cost-effective. The authors recommend integrating blockchain, SDN, and hybrid deep learning algorithms to identify vulnerabilities and attacks on IoT devices. However, the attack detection capabilities of this system remain incomplete and limited. The reliance on SDN controllers introduces a potential single point of failure, which attackers could exploit.

Benaddi et al. [17] propose Conditional Generative Adversarial Networks (GAN) to model and detect adversarial attacks in IoT networks, but it has several limitations. It does not adequately address the scalability of the proposed approach in large-scale IoT deployments with heterogeneous devices. Furthermore, the absence of real-world datasets and benchmarks limits the generalizability and practicality of the findings.

Rihan et al. [18] proposed a methodology that employed deep learning models and ensemble feature selection techniques to identify attacks on IoT networks. introduces a method combining ensemble feature selection and deep learning for attack detection in IoT networks. Despite its contributions, the approach has limitations. The reliance on deep learning models may lead to high computational overhead, making it less suitable for real-time applications on resource-constrained IoT devices. The study does not address the potential challenges of handling imbalanced datasets, which could affect detection accuracy for rare attack types. Additionally, the scalability of the proposed approach in large and diverse IoT environments with heterogeneous devices remains unexplored, limiting its applicability in real-world scenarios.

III. PROPOSED METHODOLOGY

The detection of adversarial attacks is a crucial area that has attracted substantial interest from researchers. Numerous techniques have been developed to improve the detection performance in the IoT environment for identifying these attacks. Artificial Intelligence (AI) plays a vital role in facilitating early detection, and the proposed method utilizes an AI-driven, deep-learning-based approach to identify and filter out adversarial attacks.

The proposed work combines various filtering techniques and integrates classifiers with an adaptive weight-sharing approach to filter noise, extract features efficiently, and enhance classification performance. The proposed model comprises four key steps: (A) Data Collection, (B) preprocessing, (C) Feature Extraction, and (D) classification, as illustrated in Fig. 2.

A. DATA COLLECTION

The proposed model's data collection leverages two key datasets: the open-source IoT and IIoT intrusion dataset [19] and the RT-IoT2022 dataset [20]. Both datasets, sourced from the Kaggle repository, comprehensively represent real-world scenarios, including normal and adversarial network behaviors. These datasets were selected for their robust coverage of IoT and IIoT environments, encompassing sensor data, device logs, and various measurements from IoT-enabled devices and industrial systems.

The datasets include data from various smart devices, such as wearables, home automation systems, and environmental sensors. They feature variables like temperature, humidity, motion, device interaction logs, and network traffic. These datasets serve as invaluable resources for training machine learning models, facilitating predictive analytics, and addressing challenges such as anomaly detection, optimization, and security in connected systems.

By replicating the conditions of a typical home IoT network, the datasets present a diverse mix of wired and

wireless devices, including IoT and other networked devices commonly found in residential settings. This heterogeneity provides a robust foundation for AI and data science initiatives in consumer-grade IoT and industrial IoT contexts. These datasets present numerous advantages over other models when applied to IoT and IIoT environments. Key benefits include their representation of real-world data, encompassing diverse IoT devices and scenarios. They feature a comprehensive mix of standard and adversarial data, enabling robust training and evaluation. Their large-scale nature and extensive coverage across various IoT operations further enhance their utility. Additionally, these datasets are well-suited for supporting machine learning models and facilitating the development of advanced solutions for IoT and IIoT applications. The effectiveness of the proposed model was benchmarked against existing deep-learning algorithms, showcasing its superior performance in these environments.

B. PREPROCESSING

As illustrated in Fig. 2, the preprocessing stage involved cleaning the raw dataset using standardization and normalization techniques. We introduce a three fold filtering approach, which includes Adaptive Weighted Gaussian filtering (AWGF) [21], adaptive weighted mean filtering (AWMF) [22], and adaptive wavelet filtering (AWF) [23]. Adaptive weighting mechanisms in convolutional layers offer superior performance by allowing for dynamic, context-sensitive adjustment of feature importance, leading to more flexible, efficient, and scalable models compared to standard attention mechanisms or traditional feature extraction methods. Their ability to adapt in real-time to the input data and adjust computational resources accordingly makes them highly effective for tasks that require precision and the handling of complex, varied inputs [24].

1) ADAPTIVE WEIGHTED GAUSSIAN FILTERING

High-frequency components are often filtered out using a Gaussian filter to reduce the impact of adversarial perturbations. According to Gaussian filtering, significant precision in categorization has been compromised to safeguard certain parts of the data against attacks. Structured data information can be preserved by efficiently eliminating the high-frequency components in low-pass filters such as Gaussian filtering. A Gaussian filter is represented in the time domain, and its parameters are variance (σ_g^2) and mean (μ_g).

$$Gaus(\mu_g, \sigma_g^2, t) = \frac{1}{\sqrt{2\pi\sigma_g^2}} \exp\left\{-\frac{(t-\mu_g)^2}{2\sigma_g^2}\right\} \quad (1)$$

The information above shows that measurement noise and prior process knowledge are key to Gaussian filtering accuracy. To improve the efficiency of traditional Gaussian filtering, we introduce a random weight into the above equation, and adaptive weighted Gaussian filtering

can be expressed as in (2):

$$Gaus(\mu_g, \sigma_g^2, t) = W_g \frac{1}{\sqrt{2\pi\sigma_g^2}} \exp\left\{-\frac{(t-\mu_g)^2}{2\sigma_g^2}\right\} \quad (2)$$

The expression of random weight, W_g is given in (3):

$$W_g = \frac{\|\Delta S_{k-j}\| \times \|\Delta M_{k-j}\|}{\sum_{j=1}^n \|\Delta S_{k-j}\| \times \|\Delta M_{k-j}\|} \quad (3)$$

where $\|\Delta S_{k-j}\|$ represents the state prediction, $\|\Delta M_{k-j}\|$ denotes the measurement of the residual vector.

The Gaussian filtering approach based on introduced random weights can increase the estimate precision by adaptively adjusting the measurement noise statistics and process weights to limit the impact of system noise interference on state estimation.

2) ADAPTIVE WEIGHTED MEAN FILTERING

Mean filtering is a low-pass filter that effectively removes high-frequency components while preserving the structural integrity of network data. The primary goal of adaptive weighted mean filtering is to denoise the performance by minimizing the detection errors and replacing noisy candidates. This variation in mean filtering is employed in data processing to reduce noise while maintaining the essential data characteristics. Unlike traditional mean filtering, which utilizes a fixed kernel, adaptive weighted mean filtering adjusts its kernel based on local features of the data. Adaptive weighted mean filtering can be expressed as in (4). In adaptive weighted mean filtering, each neighboring value $f(N)$ is assigned a weight $w(N)$ based on local features to enhance noise reduction.

$$M(j) = \left(\frac{\sum_{N \in S^w} (N) f(N)}{\sum_{N \in S^w} (N)} \right) \quad (4)$$

where N represents the size of the data, $w(N)$ is the weight assigned to each neighboring point N , and adaptively computed based on local data characteristics such as variance, intensity differences, or noise distribution.

3) ADAPTIVE WAVELET FILTERING

This preprocessing technique employs wavelet transformations to reduce noise and remove extraneous features from the datasets. The wavelet filter is more reliable and suitable for detecting subtle impulse-like signals that are indicative of mechanical defects. This filtering method allows for efficient extraction of meaningful information from noisy signals. Importantly, a fundamental property of the Fourier transform is that convolution in one domain corresponds to multiplication in another domain. Consequently, (5) can be restated as follows:

$$W.F(p, q) = \sqrt{p} F^{-1} \{X(f) \varphi^*(pf)\} \quad (5)$$

where F^{-1} indicates the inverse Fourier transform, $X(f)$ and $\varphi(f)$ are represented, and the Fourier transforms of $x(t)$ and $\varphi(t)$ respectively.

Equation (5) illustrates how the wavelet transform can be considered a particular filtering function. Dilating an analytical wavelet yields frequency segmentation. Alternatively, the wavelet transform convolution process is essentially a filtering mechanism. Another characteristic of (5) is that $W.F(p, q)$ quantifies the similarity between the wavelet function and signal $x(t)$, and provides information about $x(t)$ at various resolution levels. This suggests that if the wavelet is similar to the feature components concealed in the signal, it can be used for feature discovery. The critical step in the wavelet transform process, which is wavelet decomposition-based de-noising, is explained in detail below.

Multi-Resolution Analysis (MRA) is the foundation of wavelet de-noising [25]. Multi level wavelet decomposition can be used to generate the discrete details and approximation coefficients. Subsequently, the underlying patterns at various resolution levels were determined using a Discrete Wavelet Transform (DWT). Equation (6) describes the underlying model for a noisy signal:

$$x(n) = s(n) + \rho w(n) \quad (6)$$

where the noise level in this basic model is denoted by ρ ; and $w(n)$ is the typical Gaussian white noise. Recovering $s(n)$ and suppressing noise component of the signal $x(n)$ are the goals of denoising.

The DWT breaks down the data into a collection of mutually orthogonal wavelet basis functions. The mother wavelet was represented by scaled, translated, and dilated wavelet basis functions. It can be used to deconstruct data because it comes with a collection of transformations, each with a distinct set of wavelet basis functions and mother wavelets. MRA uses signal processing techniques to generate several levels of representation to understand network traffic data. Statistical validation was conducted to select important characteristics because it is essential to eliminate superfluous coefficients. The selected coefficient of the noise level (ρ) in (7) can be selected and represented as follows using MRA:

$$\rho = \{\sigma_{k'j}\}, \quad k = 1, 2, \dots, n \quad (7)$$

where n represents the length of the selected feature.

After the preprocessing stage using three-fold filtering, the pre-processed data were applied to the feature extraction process to extract the data.

C. FEATURE EXTRACTION

As shown in Fig. 2, the proposed model adopts a two level feature extraction where a modified convolution neural network (CNN) based framework is utilized in the initial phase for feature extraction. A weighted Adaptive Batch Normalization was introduced using a novel self adaptive quantum-inspired coati optimization algorithm (Q-COA) [26]. In addition, an adaptive dilated enriched convolution (ADEC) operation is performed during the CNN-based feature extraction. Q-COA's role in feature optimization is crucial [27] because it enables:

- 1) Efficient exploration of the feature space
- 2) Selection of the most relevant features
- 3) Multi-objective optimization of model performance
- 4) Reduced overfitting and improved generalization
- 5) Faster convergence and computational efficiency

A deep learning method called a Convolutional Neural Network (CNN) consists of convolutional layers that utilize various kernels to extract feature maps from images. These feature maps were then down sampled through the pooling layers. There are two main types of pooling layers: max-pooling and average-pooling. In addition, CNNs include other components, such as dense and dropout layers. The primary advantage of CNNs is their ability to automatically generate features, which makes them particularly effective for feature extraction in adversarial attack detection. The architecture of a CNN comprises input, pooling, convolutional, fully connected (FC), and output layers. The numbers of convolutional and pooling layers can vary depending on the specific topology of the CNN. The convolution process is expressed as follows.

$$C_i = f(C_{i-1} \times w_i + b_i) \quad (8)$$

A basic neural network with an input $X \in R^{P_1 \times 1}$ is considered. It consists of an FC layer with weight matrix $W \in R^{P_1 \times P_2}$ and bias $b \in R^{P_2 \times 1}$, BN layer with mean and variance ($i \in \{1 \dots p_2\}$), and 1 nonlinear transformation layer $f(\cdot)$, where P_1 and P_2 correspond to input and output feature sizes respectively. The output of this network $f(W_a x + b_a)$, where $W_a = W^T S^{-1}$, $b_a = -W^T S^{-1} \mu + b$.

The adaptive weighing mechanism in convolutional layers assigns adaptive weights to the features, allowing the model to focus on more relevant patterns and efficiently capture variations in data by adjusting weights instead of relying on static feature maps. It also reduces overfitting by automatically tuning the importance of features instead of fixed weights and also helps suppress irrelevant or noisy features, making the model more effective in addressing adversarial attacks.

1) BATCH NORMALIZATION

Batch Normalization (BN) is a layer that facilitates autonomous learning at each level of the network. It is used to standardize the outputs from the previous layers by scaling the activation of the input layer. When learning is conducted in batches, batch normalization becomes particularly effective and can serve as a form of regularization to prevent the model from overfitting. This layer can be integrated multiple times between the layers of the model, typically positioned immediately after the definition of the sequential model, and after the convolutional and pooling layers. BN can be expressed as:

$$N_B = \gamma \frac{x - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}} + \beta \quad (9)$$

In (9), the network may adjust an imposed distribution using the trainable parameters γ and β . A very small constant

for numerical stability is represented by ϵ . It is important to select a batch size that allows for the computation of mean and variance estimations. More precisely, the variance σ_j^2 and mean μ_j of adaptive Batch Normalization (AdaBN) can be updated as follows:

$$\mu_j = (1 - \alpha) \mu_{j-1} + \alpha x \quad (10)$$

$$\sigma_j^2 = (1 - \alpha) (\sigma_j^2 - 1 + \alpha (x - \mu_{j-1})^2) \quad (11)$$

where α represents a weighting factor; a new instance's effect on the running mean and variance estimation can be controlled by selecting a value for α between 0 and 1. Thus, the specified setting permits the balance of the stability-plasticity compromise.

2) ADAPTIVE BATCH NORMALIZATION

The core principle of Adaptive Batch Normalization (AdaBN) [28] uses domain-specific normalization tailored to different contexts. AdaBN aims to recalibrate the batch normalization parameters (mean and variance) during testing or deployment to align them with the new data distribution. This recalibration is crucial for scenarios involving domain shifts (e.g., models trained in one environment but deployed in another) or adversarial attacks where data properties differ from the training set. As described in Algorithm 1, the model uses traditional batch normalization to compute and store global statistics (mean and variance) for each layer based on the training dataset. Instead of using stored training statistics, AdaBN dynamically recalculates the mean and variance for each mini-batch based on the new data domain during the testing phase. A weighted factor was incorporated into AdaBN, resulting in Weighted Adaptive Batch Normalization (WAdaBN) to enhance the efficiency of batch normalization.

This enhancement is achieved through the Quantum-inspired Coati Optimization Algorithm (Q-COA). The swarm-based intelligence algorithm, COA was utilized to address the optimization challenges. The weight factor was integrated into AdaBN using Q-COA, and the weighted vectors for monitoring placements were updated based on specific criteria for each new data point. The AdaBN is described in Algorithm 1. The fitness function of Q-COA for AdaBN is expressed as in (12):

$$f(W_A) = \alpha \cdot \left(\frac{r \cdot w_1 + r \cdot w_2}{2} \right) \quad (12)$$

where w_1 and w_2 are the weighted values, r represents random numbers ranging from 0 to 1, α indicates the control parameter and value of α is 0.5. The weighted values are expressed as given follows:

$$w_1 = \frac{w_{min} - (w_{min} - w_{max})}{(w_{min} - w_{max})} \quad (13)$$

$$w_2 = \frac{w_{max} - (w_{max} - w_{min})}{(w_{max} - w_{min})} \quad (14)$$

In (14) the w_{min} and w_{max} values are set as 0.5 and 0.8, respectively. The optimal value was selected from (15),

and the obtained value was applied for batch normalization. Then, the normalization equation can be rewritten as:

$$N_B = W_A \cdot \gamma \frac{x - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}} + W_A \cdot \beta \quad (15)$$

The applied, weighted values drive the control parameters to select the optimal solution.

Algorithm 1 Adaptive Batch Normalization Input and Output

Begin

for neuron j in CNN, **do**

 Get neuron response for every data point.

 Evaluate the mean and variance for the target domain using (10)

and (11)

end for

for testing data in the target domain, **do**

 Set $w_{min} = 0.5$, $w_{max} = 0.8$, and $\alpha = 0.5$

 Compute the fitness function of QCOA using (12)

 Choose the best optimal function

 Apply selected optimal value to a batch normalization function

 Compute AdaBN using (15)

end for

end

3) ADAPTIVE DILATED ENRICHED CONVOLUTION

The Adaptive Dilated Enriched Convolution (ADEC) [29] operation was performed during the CNN-based feature extraction process. ADEC facilitates feature extraction by setting varying dilation rates for different channels, enabling the creation and fusion of multiscale features while maintaining the exact spatial dimensions. This approach allows the ADEC to modify the fused scales more effectively. In adaptive dilated convolution, the dilation rates are not manually adjusted. They are adaptively generated through a dilation rate regression module. This module comprises two fully connected layers with nonlinear activation and a global-average pooling layer. If the dilation rate module (DRM) is denoted as function $\varphi(\cdot)$, the resulting dilation rate (r) can be expressed as follows:

$$r = \varphi(x) \quad (16)$$

There are g dilation groups within input channels. Each group contained C_{in}/g channels. The dilation rates of the channels in the same group were the same. Consequently, r has the form $g \times 1$. The C_{th} input channel dilation rate was $r[c/g]$. Every channel has a different dilation rate, if $g = C_{in}$. Every channel has the same dilation rate if $g = 1$.

The Q-COA technique can also enhance feature acquisition by considering statistical characteristics [30] such as standard deviation and adaptive weighted concatenated features (including weighted mean, weighted median, weighted variance, and weighted skewness). These metrics provide insight into the structure, distribution, and central tendency of a dataset.

The total compute complexity (O) for the self-adaptive Q-COA algorithm is:

$$O = T \cdot (N \cdot (C_f + D) + N^2) \quad (17)$$

where T denotes the number of iterations, N denotes population size, C_f denotes the Cost of objective function evaluation, and D denotes the problem's Dimensionality.

Assuming $C_f = O(D)$ i.e., the fitness evaluation scales linearly with the problem dimensionality. For the dataset1 in the proposal, $D = 67$, $T = 1000$, and $N = 50$. This resulted in the computing complexity of 3,350,000 operations.

D. CLASSIFICATION

As shown in Fig. 2 and Fig. 3, the self-attention dual classification layers use a recurrent convolutional neural network (RCNN) [31] and a Restricted Boltzmann Machine (RBM) [32]. These classifiers are interconnected by adapting a self attention layer-based weight-sharing mechanism. Finally, the classifier effectively detects adversarial attacks. In addition, a predominant two-level weight-sharing mechanism enables better results than traditional classifiers such as SVM. Combining RBM and RCNN offers a unique advantage by leveraging the unsupervised feature learning capabilities of RBM with the sequential and spatial processing abilities of RCNN. This combination results in a more comprehensive model capable of handling complex datasets, reducing overfitting, improving generalization, and capturing both global and local patterns [33].

1) RECURRENT CONVOLUTIONAL NEURAL NETWORK

The characteristics of unstructured data were extracted using the RCNN method. The input, hidden, delay, and output layers comprise the RCNN. The final hidden unit, $h(t-1)$, may act as feedback to the following hidden unit, $h(t)$, until the subsequent time step is handled below owing to the delay unit. The calculation formula is as follows. If the input is $x(t)$:

$$h(t) = f(x(t), h(t-1), \theta) \quad (18)$$

where parameter θ may be obtained via the backpropagation approach and $f(\cdot)$ is the activation function. In particular, the feedforward and recursive inputs of an RNN unit are denoted by $x(t)$ and $h(t-1)$ respectively. Feedforward and recursive connections are utilized to compute the convolution in a manner that differs from the conventional CNN.

$$h(t)_i = f(w_i^f x(t) + w_i^r h(t-1) + b_i) \quad (19)$$

where b_i represents the deviation of the feature graph, $h(t)_i$ denotes the output of the feature graph and w_i^f and w_i^r are the weight matrices of the feedforward and recursive connections of the i th feature graph, respectively. The description above indicates that, after each time step, the sliding windows of the recursive connections between the convolutional layers grow continually. Fine-grained characteristics of unstructured textual data can be extracted using this technology. The Recurrent Convolutional Neural Network (RCNN) output is connected to the self-attention layer.

The self-attention layer is a valuable deep learning mechanism that allows a sequence to learn from itself. This approach enhances the model's ability to learn representations and effectively generate translations through

weight sharing. Transformer-based models utilize this self-attention layer to improve their performance. By computing the attention scores among different segments of the input sequence, the self attention mechanism helps the model to focus on the most relevant parts of the input. This enables each input to interact by determining the attention of each input relative to a specific reference input. There are several types of attention mechanisms in deep learning, with the self attention mechanism being one of the most powerful for weight-sharing. Consequently, self-attention is applied between the RCNN and the Restricted Boltzmann Machine (RBM), serving as a weight-sharing tool for interconnection.

2) SELF ATTENTION MECHANISM

The attentive vector for the convolved features can be computed after applying the attention mechanism to the series of hidden states obtained from the recurrent convolution. This model leverages an intra layer recurrent connection within a convolutional layer. Consequently, recurrent relationships allow each hidden unit h_i to influence a significant number of nearby hidden units. The attention weight of each hidden unit h_i is defined as

$$a_i = W_i * [h_i^l, h_i, h_i^r] + b_i \quad (20)$$

where h_i represents the hidden units, h_i^l and h_i^r indicate the left and right context of the hidden unit, and all concealed states (convolve features) are used as inputs in self-attention to obtain the attentive vectors. Consequently, this technique uses a weighted sum over all convolved features to obtain the self-attentive feature vector C_i for all h_i .

$$C_i = \sum_i \text{Softmax}(a_i) \quad (21)$$

where W_i denotes the weight matrix and a_i is the attention weight. The RCNN is convolved with this self-attention to produce an attention weight. This weight was shared with the RBM using a self-attention mechanism.

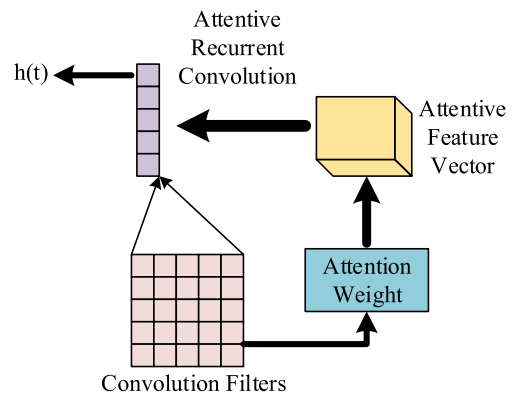


FIGURE 3. Structure of self-attention layer.

3) RESTRICTED BOLTZMANN MACHINE (RBM)

Using a hidden layer of binary stochastic units, $h = (h_1, \dots, h_H)$, the RBM joint distribution of an input,

$x = (x_1, \dots, x_D)$, and a target class, $y \in \{1, \dots, C\}$ is classified. To accomplish this, we first define an energy function.

$$E(y, x, h) = X \quad (22)$$

where $X = -h^T W_x - b^T x - c^T h - d^T e_y - h^T U e_y$

Based on the energy function, the following probability determines the values of x , y and h :

$$p(y, x, h) = \frac{\exp(-E(y, x, h))}{Z} \quad (23)$$

where Z is the normalization constant, ensuring that (23) is a valid probability distribution. The necessary conditional distributions were straightforward. It may be expressed as follows when conditioning on the visible layer:

$$p(h|y, x) = \prod_j p(h_j|x, y), \quad \text{with} \\ p(h_j = 1|y, x) = \text{sigm}\left(c_j + U_{jy} + \sum_i W_{ji}x_i\right) \quad (24)$$

where,

$$\text{sigm}(a) = 1/(1 + \exp(-a)) \quad (25)$$

When conditioning on the hidden layer, it can be expressed as follows:

$$p(x|h) = \prod_i p(x_i|h), \quad \text{with} \\ p(x_i = 1|h) = \text{sigm}\left(b_i + \sum_j W_{ji}h_j\right) \quad (26)$$

$$p(y|h) = \frac{\exp\left(d_y + \sum_j U_{jy}h_j\right)}{\sum_{y^*} \exp\left(d_{y^*} + \sum_j U_{jy^*}h_j\right)} \quad (27)$$

It is also feasible to conduct classification by precisely computing $p(y|h)$. Finally, the classifier can effectively detect adversarial attacks using the RBM. In addition, a predominant two-level weight-sharing mechanism provides better results than a traditional classifier.

IV. RESULTS AND DISCUSSION

The results are evaluated in the following section. Various metrics [34] are used to assess the effectiveness of the proposed strategy. The primary purpose of the metrics is to compare the effectiveness of the recommended strategy with other existing algorithms, such as RBM, RCNN, GRU, and DCNN. The key objectives of this study were as follows:

- 1) The effectiveness of the proposed method is evaluated using ROC (Receiver Operating Characteristic) and AUC (Area Under the Curve), which are the metrics commonly used to assess the performance of models.
- 2) Evaluated the confusion matrix, which provides essential insight into a model's classification performance for an optimal threshold specific to the model.
- 3) Jupyter Notebook is the platform used to experiment with AI models.

The following Evaluation Metrics [34] were used in assessing the detection of adversarial attacks and defenses in an IoT environment:

$$TPR = \frac{TP}{TP + FN} \quad (28)$$

$$FNR = \frac{FN}{FN + TP} \quad (29)$$

TP, True Positive, TN, True Negative, FP, False Positive, and FN, False Negative.

where TPR measures the proportion of actual positives (true positive cases) correctly identified by the model, and FPR measures the proportion of actual negatives (true negative cases) that were incorrectly classified as positive by the model.

TPR shows how good the model is at detecting positive instances. FPR shows how often the model incorrectly labels negative instances as positive. ROC and AUC are derived from the relationship between the False Positive Rate (FPR) and True Positive Rate (TPR). The ROC curve is a graphical representation of the trade-off between TPR and FPR at various classification thresholds. At the same time, the AUC quantifies the overall ability of the model to distinguish between classes, with higher values indicating better performance.

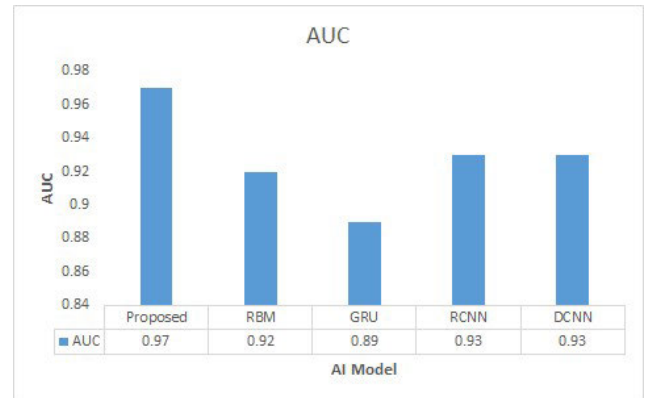


FIGURE 4. Receiver Operating Characteristic (ROC) – dataset [18].

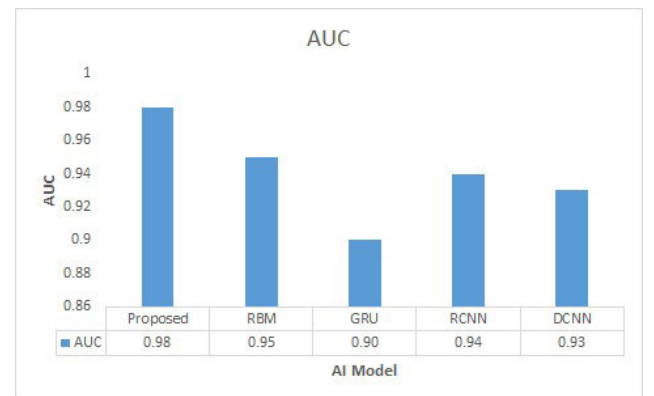


FIGURE 5. Receiver Operating Characteristic (ROC) – dataset [19].

The performance of the proposed model was evaluated by comparing it with several existing standalone methods,

TABLE 2. Hyperparameters and training details.

Parameter	RBM [32]	RCNN [31]	GRU [32]	DCNN [32]	Proposed Model
Input shape	(83,)	(83, 1)	(83, 1)	(83, 1)	(83, 1)
Input layer	Dense (83, Sigmoid)	Conv1D (32 filters, 3 kernel size, ReLU)	GRU (128, Tanh)	Conv1D (32 filters, 3 kernel size, ReLU)	Three-Stage Filtering with Adaptive Weights
Pooling 1	--	MaxPooling1D (2 pool size)	--	MaxPooling1D (2 pool size)	Adaptive Feature Extraction (Dilated Enriched Convolution)
Hidden Layer 1	Dense(128, Sigmoid)	Conv1D (64 filters, 3 kernel size, ReLU)	GRU (128, Tanh)	Conv1D (64 filters, 3 kernel size, ReLU)	RCNN + RBM with Self-Attention Mechanism
Pooling 2	--	MaxPooling1D (2 pool size)	--	MaxPooling1D (2 pool size)	Multi-Scale Adaptive Pooling
Hidden Layer 2	Dense (64, Sigmoid)	LSTM (128, Tanh)	Dense (128, ReLU)	Dense (128, ReLU)	Quantum-Inspired Coati Optimization for Feature Selection
Output Layer	Dense (12, Softmax)	Dense (12, Softmax)	Dense (12, Softmax)	Dense (12, Softmax)	Dense (12, Softmax) (Q-COA Optimized)
Learning Rate	0.001	0.001	0.001	0.001	Adaptive (Dynamic Adjustment via Q-COA)
Activation Functions	Relu, Sigmoid	Relu, Sigmoid	Relu, Sigmoid	Relu, Sigmoid	ReLU, Self-Attention (RBM & RCNN), Sigmoid
Optimizer	Adam	Adam	Adam	Adam	Q-COA Optimized Adaptive Adam
Batch Size	32	32	32	32	32 (Adaptive)
Epochs	300	300	300	300	300
Early Stopping	Yes	Yes	Yes	Yes	Yes
Patience	10	10	10	10	10
Restore Best Weights	Yes	Yes	Yes	Yes	Yes
Dropout rate	--	0.5	--	0.5	Adaptive
Strides	--	1	--	1	Adaptive
Max pooling	--	2x2	--	2x2	Adaptive
Padding	--	Same	--	Same	Adaptive

including RCNN, GRU, RBM, and DCNN [32]. These models were implemented in Python, trained, and tested using the datasets [18], [19]. A standard 80/20 data split was applied, with dynamic validation employed during training. Validation data was extracted from the training set to monitor the model's performance. Table 2 lists the hyperparameters for each model. Hyperparameters were fine-tuned using Early Stopping, which halts training when the model's performance on the validation set ceases to improve. In the proposed model the adaptive feature extraction mechanism includes dilated convolutions, which help expand the receptive field of the model without reducing the sequence length too quickly. This is a more complex approach, but it is very effective. The idea is to preserve or adaptively modify the receptive field, which would lead to better feature extraction for sequences. The methodology used across all the models is consistent, and the outcomes were compared with the proposed model, as illustrated in Fig. 4 and Fig. 5.

The True Positive Rate (TPR) and False Positive Rate (FPR) at the optimal threshold values for each model across the datasets are presented in Table 3 and Table 4. In Table 3, the optimal threshold values for the RBM, RCNN, GRU, DCNN, and proposed models are 0.4, 0.45, 0.41, 0.4, and 0.36, respectively. In Table 4, the optimal threshold values for these models are 0.52, 0.47, 0.6, 0.6, and 0.45, respectively. The TPR and FPR values clearly compare model performance, helping to identify the effectiveness of

the proposed model. The Proposed Model has the highest accuracy, precision, and F1-score, showing its effectiveness compared to other models, as shown in Table 3 and Table 4.

TABLE 3. Proposed method vs existing techniques (dataset [18]).

	RBM [32]	RCNN [31]	GRU [32]	DCNN [32]	Proposed Model
TPR	0.90	0.90	0.81	0.88	0.95
FPR	0.18	0.15	0.18	0.16	0.10
Accuracy	0.86	0.87	0.81	0.86	0.92
Precision	0.80	0.84	0.80	0.83	0.89
Sensitivity	0.90	0.90	0.81	0.88	0.95
F1-score	0.85	0.87	0.81	0.85	0.91

TABLE 4. Proposed method vs existing techniques (dataset [19]).

	RBM [32]	RCNN [31]	GRU [32]	DCNN [32]	Proposed Model
TPR	0.86	0.90	0.81	0.82	0.97
FPR	0.13	0.15	0.18	0.09	0.07
Accuracy	0.87	0.87	0.82	0.87	0.95
Precision	0.86	0.83	0.80	0.90	0.92
Sensitivity	0.86	0.90	0.81	0.82	0.97
F1-score	0.86	0.87	0.81	0.86	0.95

As shown in Fig. 4 and Fig. 5, the AUC for the proposed model is higher than others. This highlights the effectiveness of the proposed method in detecting and defending against adversarial attacks, attributed to its adaptive feature

extraction, noise filtering, Q-COA-based convolution, and dual-system classification approach, along with the adaptive weights.

The results demonstrate that the proposed model performs effectively due to its heterogeneous nature and the use of adaptive weights. It surpasses standalone models by leveraging the strengths of diverse models while dynamically adjusting their contributions based on robustness. This combination of diversity, adaptability, and redundancy provides a robust defense against the vulnerabilities often exploited by adversarial attacks. The following factors contribute to the superior performance of the proposed method:

- 1) **Enhanced Diversity:** The method utilizes a variety of models, each with unique strengths and weaknesses. This diversity ensures that adversarial perturbations affecting one model are less likely to impact others similarly. As a result, the overall robustness of the system improves through this complementary approach.
- 2) **Redundancy:** The proposed approach incorporates redundancy, ensuring that even if an adversarial attack compromises one model, other models can still contribute to accurate classification. This reduces the likelihood of a successful attack, as adversaries must simultaneously fool multiple models.
- 3) **Performance-Based Weighting:** Adaptive weights prioritize models that demonstrate better performance on specific instances or subsets of data. Under adversarial conditions, models trained with techniques such as adversarial training or noise robustness are given more significant influence, enhancing the models' effectiveness in mitigating attacks.

These characteristics collectively make the proposed model an effective solution against adversarial attacks.

The proposed model has slower training speed, slower inference, and higher GPU memory usage to store both feature representations and sequential state information. The Proposed model will have a higher computational overhead, as it combines the benefits of RBM (for feature extraction) and RCNN (for sequential processing), but its superior performance in terms of the evaluation metrics makes it a strong candidate in practical applications. However, these performance benefits come at the cost of increased inference time, training speed, and GPU memory usage.

V. CONCLUSION

This paper presents a novel deep-learning approach for detecting adversarial attacks within IoT environments by integrating advanced machine-learning techniques to enhance detection accuracy and efficiency. This approach filters and mitigates adversarial attacks locally at the edge, reducing computational reliance on cloud-based solutions, which can be computationally intensive and costly. The proposed method is also useful in the cloud to address higher-order adversarial attacks at the network level.

The model introduces a dual-classification framework and a two-tier feature learning strategy to improve classification precision, preprocessing effectiveness, and feature extraction capabilities. In the preprocessing phase, a three-step filtering process comprising adaptive wavelet filtering, adaptive weighted mean filtering, and adaptive weighted Gaussian filtering was employed to eliminate high-frequency noise, optimize denoising performance, and remove irrelevant features from the datasets.

A two-level approach leveraging a modified Convolutional Neural Network (CNN) architecture was utilized for feature extraction. This method incorporates adaptive dilated enriched convolution operations along with a self-adaptive Quantum-inspired Coati Optimization Algorithm (Q-COA) with Weighted Adaptive Batch Normalization (AdaBN), aiming to optimize statistical feature properties, including adaptive weighted concatenated features and standard deviation.

A dual-classification system composed of a Recurrent Convolutional Neural Network (RCNN) and a Restricted Boltzmann Machine (RBM) was employed in the classification phase. These models are interconnected via a weight-sharing mechanism supported by a self-attention layer, which strengthens classification performance by implementing a two-level weight-sharing mechanism that effectively identifies and filters adversarial attacks.

Evaluation of the proposed model revealed superior performance compared to existing methods, as demonstrated through key metric AUC. The model consistently achieved higher AUC, surpassing alternative approaches. This study provides an innovative solution for detecting and mitigating adversarial attacks in IoT systems, showcasing the model's robust performance. The proposed model is versatile and is suitable for applications beyond IoT-specific contexts.

This paper proposes a methodology to address adversarial attacks in IoT environments, but several challenges must be considered for real-world IoT applications. One of the primary concerns is the constraints on computing and memory availability, which are crucial for the efficient operation of IoT systems. To address this issue, the AI models should run on hardware accelerators [35] rather than on the processor. Additionally, while the latencies introduced by the proposed methodology are slightly higher than single model approaches, they do not significantly impact the response time requirements of IoT devices at the edge, which typically operate within a few microseconds to milliseconds range. However, systems such as autonomous vehicles or industrial control systems, which require low-latency responses, necessitate careful planning of these latencies during the design phase. Furthermore, IoT devices face the challenge of pretraining AI models, which requires accurate characterization of the devices to ensure reliable performance in the field. Unfortunately, obtaining such characterization data can be expensive, thereby increasing the overall cost of IoT products. In addition, IoT devices are vulnerable to noise

and environmental interference, which complicates the detection of adversarial attacks. Finally, the adversarial defense mechanisms incorporated in the proposed methodology may increase energy consumption, which poses a significant challenge for battery-powered IoT devices. These concerns are particularly relevant for IoT devices operating at the edge, where resource constraints are more pronounced. Every IoT design is distinct and must be characterized and trained using data specific to its unique architecture.

Although the proposed solution demands substantial computational resources and power, training AI models is a computationally intensive process, therefore, pre-training these models before deploying them on IoT edge devices is a more practical approach. The computing resources at the edge can be carefully configured to efficiently process the data generated locally. Furthermore, advancements in edge computing facilitate the efficient management of edge devices [36], optimize resource utilization, and integrate low-power methodologies to improve performance.

It is important to note that using AI models to mitigate adversarial attacks is unlikely to pose ethical concerns in IoT environments. It provides considerable advantages in the broader context. This work paves the way for developing comprehensive end-to-end IoT solutions that leverage AI models to mitigate adversarial attacks, thereby enhancing the overall security of IoT environments.

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